Proposal Defense

Attention-Guided Deep Learning Architecture for Camouflaged Image Segmentation

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Course: RBB4032 Final Year Project I

Cluster: Intelligent Signal and Imaging

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OUTLINE OF PRESENTATION

1. Introduction
2. Problem Statement, Objectives, Scope of Study
3. Literature Review
4. Proposed Methodology
5. Gantt Chart & Milestones
6. Preliminary Results
7. Conclusion

INTRODUCTION

Camouflaged object segmentation is challenging because objects blend with their surroundings.

Key challenges:

- **Low foreground–background contrast**
- **Weak and ambiguous boundaries**
- **Texture similarity**
- **Background-dominant pixels**
- **Occlusion and complex shapes**



Military camouflage

Generic CNNs (e.g., U-Net) struggle due to lack of camouflage-aware modeling.

Camouflage-specific models like SINet improve localization and refinement, but boundary precision and background suppression remain challenging in complex military scenes.

PROBLEM STATEMENT

1



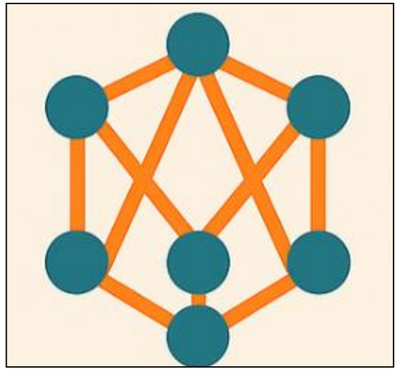
Low foreground–background contrast and high texture similarity

2



Weak boundary representation and complex environmental blending

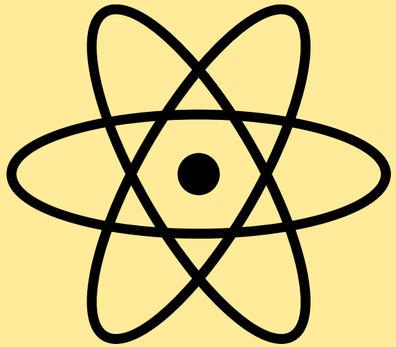
3



Existing models (e.g., U-Net and SINet) lack sufficient attention-based refinement

OBJECTIVES

1



To implement and evaluate baseline segmentation models (U-Net and SINet) for camouflaged image segmentation.

2



To develop an attention-guided enhancement integrated into SINet to improve feature discrimination and boundary precision.

3



To evaluate and compare baseline and proposed models using standard segmentation metrics (e.g., Dice coefficient and IoU).

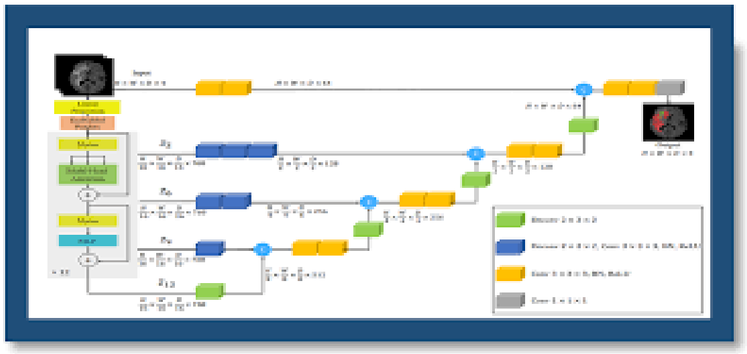
SCOPE OF STUDY

1



Focus on Malaysian military camouflage dataset for binary semantic segmentation (soldier vs background).

2



Implement baseline models (U-Net and SINet) for performance comparison.

3



Develop an attention-guided SINet-based model to improve boundary precision and feature discrimination, evaluated using Dice score

LITERATURE REVIEW

Semantic Segmentation Architectures for Camouflage

Model	Year	Core Architecture Idea	Camouflage-Specific Mechanism	Boundary Handling	Strength	Limitation	Ref
U-Net	2015	Encoder–decoder with skip connections	None (generic segmentation)	Implicit via decoder	Lightweight, works with small datasets	Struggles with low contrast & camouflage blending	[1]
SINet	2020	Two-stage Search → Identification pipeline	Search Attention (SA) to localize camouflage regions	Limited refinement	Strong first camouflage-specific baseline	Boundary precision still limited	[2]
SINet-V2	2022	TEM + Ne… Decoder … Attention…	…efi… …ixe… …ss…				…]
PFNet	2021	Positionir… Module	…rac…				…]
BGNet	2022	Edge-awa… integratic…	…nc… …br…				…]

ResNet + RF
- Extract image features
- Capture different scales

Search Attention
- Finds possible camouflage area
- Rough localization

Two-Stage Process
- First: Search
- Second: Refine
- Better than normal U-Net

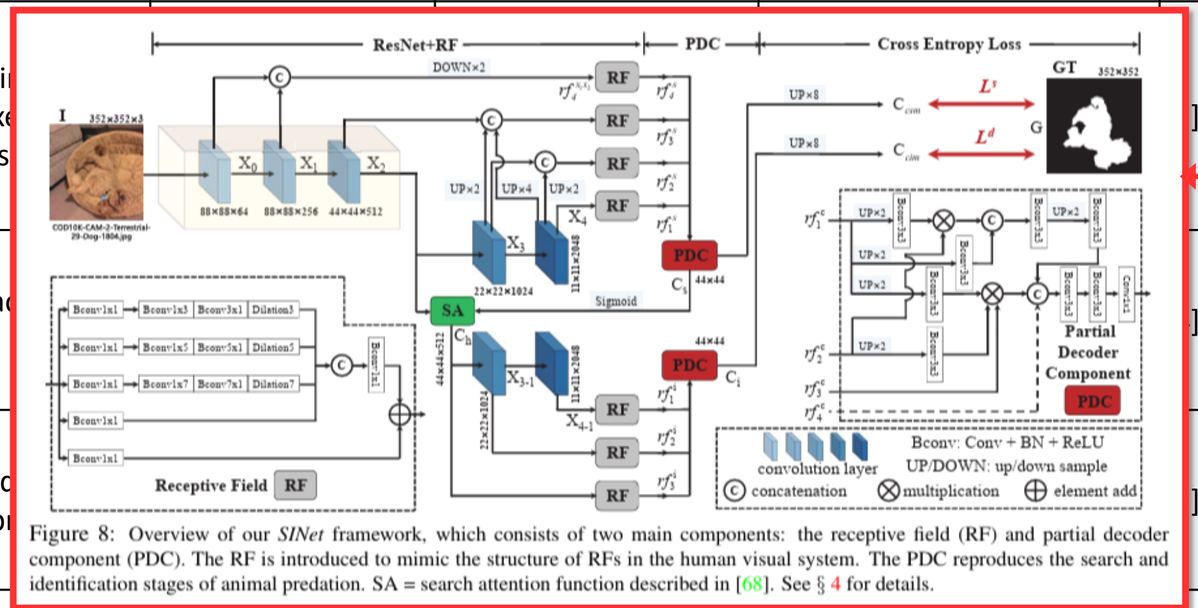


Figure 8: Overview of our *SINet* framework, which consists of two main components: the receptive field (RF) and partial decoder component (PDC). The RF is introduced to mimic the structure of RFs in the human visual system. The PDC reproduces the search and identification stages of animal predation. SA = search attention function described in [68]. See § 4 for details.

LITERATURE REVIEW

Semantic Segmentation Architectures for Camouflage

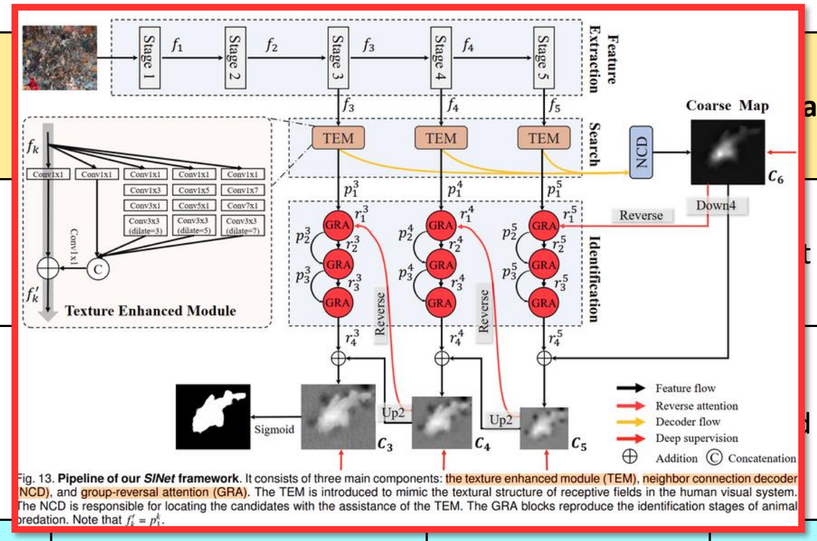


Fig. 13. **Pipeline of our *SINet* framework**. It consists of three main components: the texture enhanced module (TEM), neighbor connection decoder (NCD), and group-reversal attention (GRA). The TEM is introduced to mimic the textural structure of receptive fields in the human visual system. The NCD is responsible for locating the candidates with the assistance of the TEM. The GRA blocks reproduce the identification stages of animal predation. Note that $f_k' = p_1^k$.

Texture Enhanced Module (TEM)
- Enhances texture features
- Improves camouflage discrimination

Neighbor Connection Decoder (NCD)
- Generates coarse localization map
- Connects high-level features efficiently

Group-Reversal Attention (GRA)
- Refines detected regions
- Focuses on hard and confusing areas
- Improves boundary precision

Three-Stage Strategy
- Feature Extraction
- Search (Coarse Map)
- Identification (Refinement)

Model	Year						
U-Net	2015						
SINet	2020						
SINet-V2	2022	TEM + Neighbor Connection Decoder (NCD) + Group-Reversal Attention (GRA)	Cascaded refinement with hard-pixel weighted loss	Improved detail recovery via GRA	Most aligned with attention-guided refinement	Heavier than basic SINet	[3]
PFNet	2021	Positioning Module + Focus Module	Explicit distraction mining	Progressive ambiguous-region refinement	Strong foreground vs background separation	416×416 input; heavier training setup	[4]
BGNet	2022	Edge-aware + Edge-guided feature integration	Explicit boundary supervision branch	Dice loss on edges + weighted mask loss	Strongest boundary-focused design	Additional edge branch increases complexity	[5]

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Edge-Aware Module (EAM)
- **Extracts boundary information**
- **Detects object edges**

Edge-Guided Feature Module (EFM)
- **Uses edge information**
- **Enhances segmentation features**

Context Aggregation Module (CAM)
- **Combines multi-scale features**
- **Improves region understanding**

Boundary Supervision
- **Uses Dice loss for edges**
- **Uses BCE + IoU for mask**

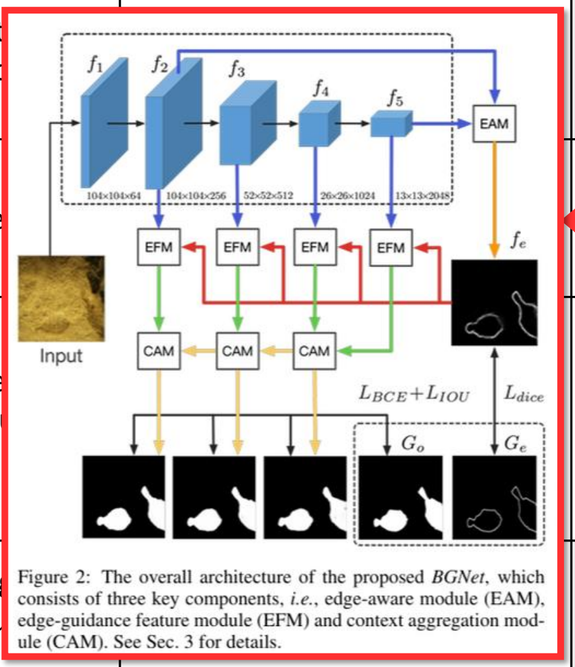


Figure 2: The overall architecture of the proposed *BGNet*, which consists of three key components, *i.e.*, edge-aware module (EAM), edge-guidance feature module (EFM) and context aggregation module (CAM). See Sec. 3 for details.

LITERATURE REVIEW

Training Techniques for Camouflage Semantic Segmentation

Technique Name	Description	Purpose (Imbalance / Boundary / Generalization)	Typical Loss Functions	Data Augmentation Strategies	Computational Cost	Pros / Cons	Ref
Transfer Learning (Pretrained Backbone)	Uses ImageNet-pretrained backbone (e.g., ResNet-50) and fine-tunes on camouflage dataset.	Generalization	BCE, BCE+Dice	Random flip, resize, crop	Low–Medium	-Faster convergence and works with small dataset -May retain bias from natural images	[2]
Hybrid Loss (BCE + Dice / IoU)	Combines pixel-wise loss and overlap-based loss to improve segmentation quality.	Class imbalance	BCE + Dice / IoU	Standard augment		-Handles background dominance and	
Coarse-to-Fine Supervision (Multi-Stage Training)	Two-stage architecture: first coarse localization, then boundary refinement (e.g., SINet).	Boundary refinement	Multi-output BCE	Multi-scale training	Medium	structured feature learning -Higher complexity	[2]
Attention-Based Feature Reweighting	Applies spatial/channel attention to suppress background and enhance foreground discrimination.	Background suppression + Boundary	BCE + attention-guided loss	Random crop, color jitter	Medium–High	-Focuses on confusing regions -Improves camouflage discrimination -Increased training cost	[7]
Semi-Supervised Learning (Pseudo-Labeling)	Uses labeled data to generate pseudo labels for unlabeled images to expand training set.	Generalization + Data scarcity	BCE + consistency loss	Strong–weak augmentation	Medium	-Utilizes unlabeled data and improves robustness -Risk of noisy pseudo-labels	[8]

Pretrained ResNet-50 backbone is used to provide strong initial feature representation, improving training stability, convergence speed, and generalization in complex camouflage environments.

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Weak boundary representation remains a key challenge even in camouflage-specific models

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Semi-supervised learning helps utilize unlabeled camouflage image (future direction)

LITERATURE REVIEW

Attention & Feature Reweighting

Why Important for Camouflage	Ref	Module	Type	What it Does
Suppresses background-dominant channels and enhances discriminative cues	[9]	**SE (Squeeze-Excitation)**	Channel Attention	Learns importance of each channel and rescales features
Low overfitting risk; suitable for small camouflage datasets	[10]	**ECA (Efficient Channel Attention)**	Lightweight Channel Attention	Simplified channel reweighting with minimal parameters
Improves boundary precision and reduces false positives	[2]	**Sinet-V2 GRA**	Reverse-Guided Refinement	Uses prediction map to guide iterative refinement

SE / ECA (Feature Recalibration)
- **Reweights feature channels**
- **Suppresses background noise**
- **Lightweight and stable**

SINet-V2 GRA (Refinement Attention)
- **Reverse-guided refinement**
- **Multi-stage structure improvement**
- **Enhances boundary accuracy**

LITERATURE REVIEW

RESEARCH GAP



- Generic segmentation models (e.g., U-Net) struggle with low contrast and camouflage blending.

- Camouflage-specific models (e.g., SINet, SINet-V2) improve localization but still face:
  > Weak boundary precision
  > Background-dominant feature activation
  > Limited early feature recalibration

- Lightweight attention integration within camouflage pipelines remains under-explored.

JUSTIFICATION



This project proposes an Attention-Enhanced SINet Framework that:

- Integrates lightweight channel attention (SE / ECA) to suppress background-dominant features.
- Enhances refinement using camouflage-specific mechanisms (e.g., GRA-style refinement).
- Improves boundary precision and discrimination in complex military camouflage scenes.
- Maintains computational efficiency using pretrained ResNet-50 backbone.

PROPOSED METHODOLOGY

Phase 1: Data Preparation

- Total Dataset (STRIDE): 500
- Image resizing (256×256)
- Binary mask conversion
- Data augmentation (horizontal/vertical flip, rotation, brightness & contrast adjustment, resizing) is applied to improve generalization.
- Train / Validation / Test split





The figure above shows sample datasets received from STRIDE.

Phase 2: Baseline Model Implementation

Baseline 1 – U-Net
- Encoder–decoder structure
- BCE / BCE + Dice loss

Baseline 2 – SINet
- Pretrained ResNet-50 backbone
- Search → Identification pipeline

Phase 3: Proposed Attention-Enhanced SINet

- Insert SE / ECA after RF feature extraction
- Integrate refinement mechanism (GRA-style)
- Fine-tune using transfer learning

Phase 4: Training Strategy

- Transfer learning (ImageNet pretrained weights)
- Hybrid loss (BCE + Dice / IoU)
- Batch size: 2–4
- 3–5 initial epochs (baseline validation)
- Early stopping

Framework Used:
PyTorch (Deep Learning framework), Torchvision (Pretrained models)

Phase 5: Evaluation

- Dice Coefficient
- Intersection over Union (IoU)
- Visual qualitative comparison

💻 Software Environment:

- Python 3.x
- PyTorch
- OpenCV
- Matplotlib
- VS Code / Google Colab

🖥️ Hardware
- NVIDIA MX350 GPU (2GB VRAM) (initial experiments)
- Optional higher-end GPU (Supervisor workstation / lab GPU)
- Windows OS

GANTT CHART & MILESTONES

FYP 1

Activities	Timeline (Week)													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Project Initiation														
Project topic selection and a brief discussion of the project details with the supervisor.														
Received dataset from STRIDE (project collaborator)														
Literature Review														
Study of semantic segmentation methods														
Study of camouflaged object detection (COD) architectures														
Study of transfer learning techniques														
Review of attention mechanisms (SE, ECA, GRA)														
Review of software tools, hardware, and system requirements														
Data Preparation and Baseline Implementation														
Dataset annotation using CVAT														
Dataset cleaning and preprocessing														
Baseline U-Net implementation														
Preliminary training and evaluation														
Baseline Implementation and Dataset Annotation														
Baseline SINet implementation														
Complete all datasets annotation														

FYP 2

Activities	Timeline (Week)													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Deep Learning Model														
Complete training and evaluation of baseline (U-Net and SINet)														
Attention module integration (SE / ECA)														
Training and hyperparameter tuning														
Refinement enhancement (Optional GRA-style)														
Final model training														
Complete attachment enhance														
GUI Simple Web Interface (Streamlit / Flask) (Optional)														
Results Analysis & Discussion														
Evaluation (Dice, IoU, Visual Analysis)														
Ablation study (optional)														

Key Milestones

PRELIMINARY RESULTS

Data Validation

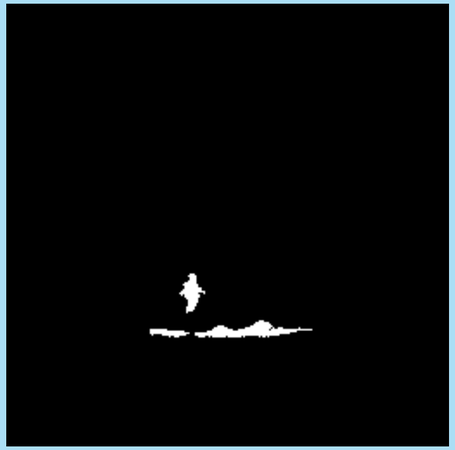
```
torch.Size([3, 256, 256]) torch.Size([1, 256, 256])
tensor([0., 1.])
Image tensor shape: torch.Size([1, 3, 256, 256])
Mask tensor shape: torch.Size([1, 256, 256])
Mask unique values: tensor([0., 1.])
```

Training performance

```
PS C:\Users\LENOVO\OneDrive - Universiti Teknologi PETRONAS\Pictures\Desktop\fyp_military> python train_unet_quick.py
Using device: cuda
Total: 96 | Train: 67 | Val: 14 | Test: 15

Epoch 7/10 | Train Loss: 1.1635 | Val Loss: 1.1644 | Val Dice: 0.2581
 Saved best model (by Dice): checkpoints\unet_best.pth
Epoch 8/10 | Train Loss: 1.1493 | Val Loss: 1.1505 | Val Dice: 0.2308
Epoch 9/10 | Train Loss: 1.1291 | Val Loss: 1.1118 | Val Dice: 0.1577
Epoch 10/10 | Train Loss: 1.1024 | Val Loss: 1.1864 | Val Dice: 0.2074
 Saved loss curve: checkpoints\unet_loss_curve.png
```

Ground Truth

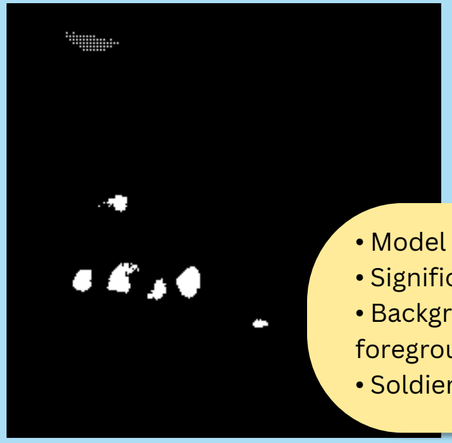


- Manual annotation from CVAT
- White = soldier (foreground)
- Black = background

Original Image

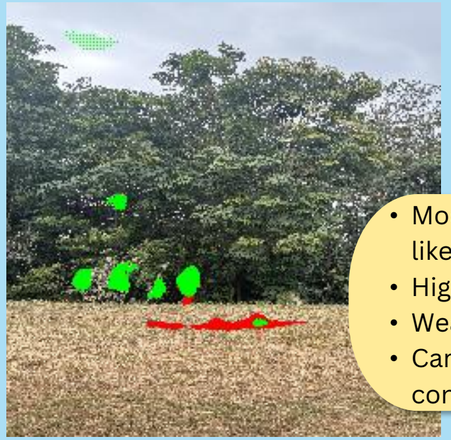


- Soldier blends with leaves
- Similar color (green/brown)
- Weak boundary between soldier and background



Prediction (Binary Mask)

- Model predicts multiple foreground regions
- Significant false positive detected
- Background regions misclassified as foreground
- Soldier region partially detected in some cases



Overlay (Red = Ground Truth, Green = Prediction)

- Model detects some object-like regions
- High false positive rate
- Weak boundary precision
- Camouflage texture causes confusion

Baseline U-Net demonstrates learning capability but struggles with background suppression and boundary refinement in camouflage environments.

CONCLUSIONS

In conclusion, research has demonstrated that generic segmentation models struggle in camouflage scenarios due to low contrast and poor boundary separation. Attention-guided and refinement-based architectures have been shown to significantly improve performance. Hence, this project proposes an attention-enhanced SINet-based framework to improve camouflaged object segmentation.

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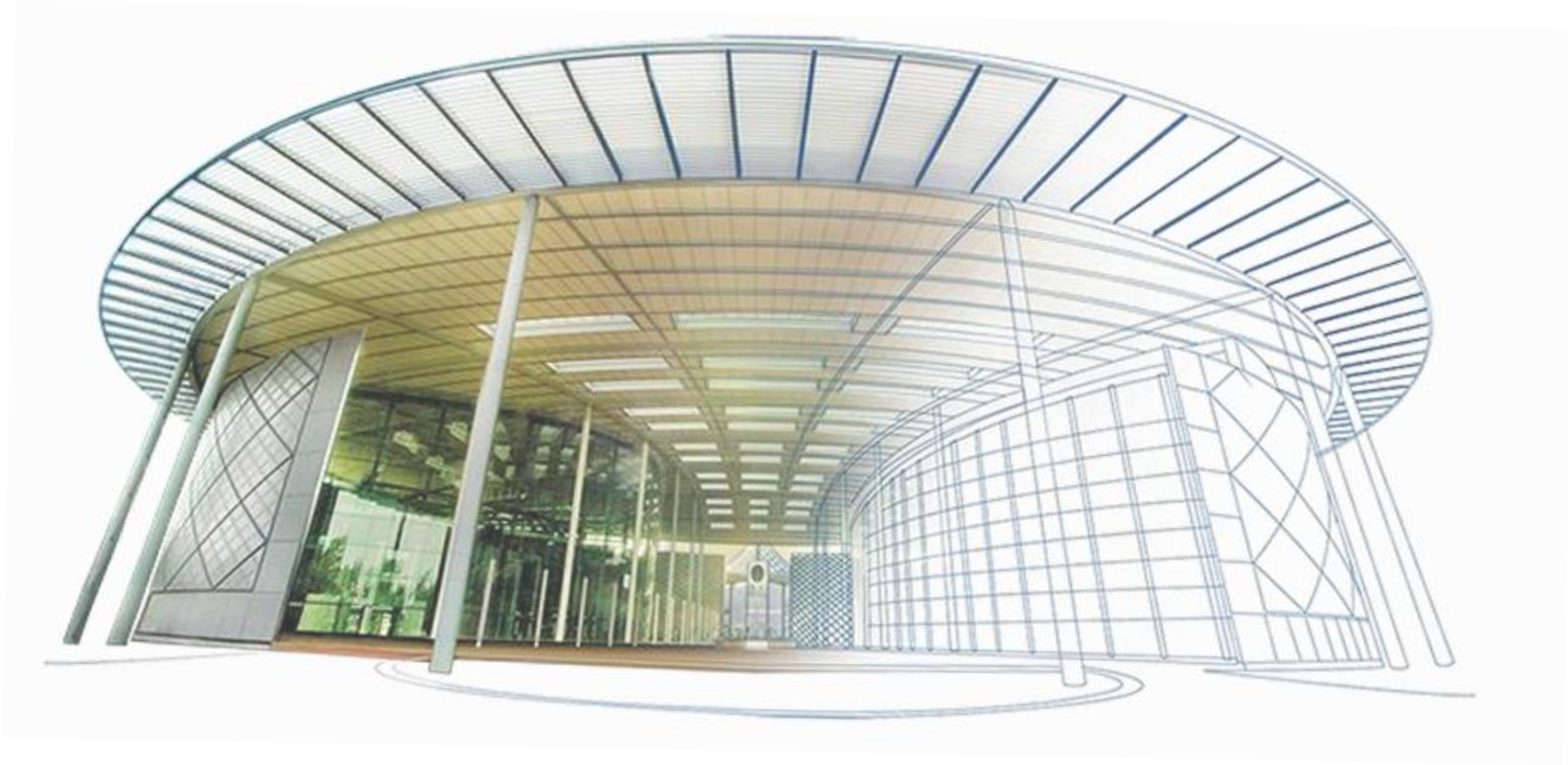
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THANK YOU

Q & A